



Figure 1: Logging system at TCE data centre

different sub-systems could be different.

- Distributed systems have different log files, all of which typically need to be 'digested' to analyse errors.
- If the log files are located within the system, then a critical error might cause the loss of log files, and hence an LMS that is physically separate from the system is needed.

The sheer size or the rate at which logs are generated mandates automation of log management, as it is very difficult to analyse rolling logs by human analysts.

There are several open source tools (like Logalyze, Greylog2, Fluentd and Logstash) that can help us in setting up an LMS. In this article, we present the details of an LMS that we have set up in the data centre at Thiagarajar College of Engineering (TCE) Madurai, Tamilnadu, using Logstash, Elasticsearch and Kibana.

The LMS workflow

Figure 1 depicts the various components and their interactions in the LMS that we built in TCE's data centre (Figure 2).

Setting up a centralised LMS using Logstash

Logstash was developed by Jordan Sissel and in one of the talks at Puppetconf, he remarked that its main objective was simplicity. Based on our experience with Logstash, it truly is extremely simple to use and also scales well with the size of the log files. Kibana is the front-end interface for Logstash that facilitates log visualisation and exploration. Logstash has a back-end to store, track and search for the desired patterns and to analyse log entries. It has a plethora of other



Figure 2: TCE data centre

options to set up an LMS. It has a graphical search console, which is depicted in Figure 3.

The Logstash pipeline

The Logstash pipeline has three phases—input, filter and output. A single Logstash instance can receive inputs/events from multiple input sources; filters are set up to collect events of interest and multiple filters can be set up to enhance the precision with which events of interest can be found. The events which pass through these filters can be sent to multiple outputs.

Input: Logstash has a forwarder component, using which log entries can be collated from multiple sources like Stdin, Files, RabbitMQ, Email, Twitter, etc. The Logstash forwarder currently supports over 40 sources. The forwarder streams log entries into Logstash, and adds *type* and some additional tags like the *host name* and forwarded timestamp to the entries. The tags help in identifying the source from which the log entry is streamed and cannot



Figure 3: The Logstash search console